1 I

1.1 (1)

The ground state satisfies:

$$a\phi_0(x) = 0.$$

Using the definition of the annihilation operator:

$$\sqrt{\frac{m\omega}{2\hbar}}\left(x + \frac{\hbar}{m\omega}\frac{d}{dx}\right)e^{-\lambda x^2} = 0.$$

Therefore:

$$x - \frac{2\hbar\lambda}{m\omega}x = 0,$$

which gives:

$$\boxed{\lambda = \frac{m\omega}{2\hbar}}.$$

Since the Hamiltonian can be written as

$$H = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right),$$

the ground state energy is:

$$\boxed{\epsilon_0 = \frac{\hbar\omega}{2}}.$$

1.2 (2)

The first excited state is obtained by applying the creation operator to the ground state:

$$\begin{aligned}\phi_1(x) &= a^\dagger\phi_0(x)\\ &= \sqrt{\frac{m\omega}{2\hbar}}\left(x - \frac{\hbar}{m\omega}\frac{d}{dx}\right)e^{-\lambda x^2}\\ &= 2\sqrt{\frac{m\omega}{2\hbar}}xe^{-\frac{m\omega}{2\hbar}x^2}.\end{aligned}$$

Since the normalization is not required, we can write:

$$\boxed{\phi_1(x) \propto xe^{-\frac{m\omega}{2\hbar}x^2}}.$$

The corresponding energy eigenvalue is:

$$\boxed{\epsilon_1 = \frac{3}{2}\hbar\omega}.$$

2 II

2.1 (3)

The two-dimensional Hamiltonian can be separated into two independent one-dimensional harmonic oscillators:

$$H = H_x + H_y.$$

Therefore, its eigenfunctions can be written as products of the eigenfunctions of the two oscillators:

$$\phi_{n_x,n_y}(x,y) = \phi_{n_x}(x)\phi_{n_y}(y).$$

The corresponding energy eigenvalues are:

$$E_{n_x,n_y} = \hbar\omega(n_x + n_y + 1).$$

Thus, the lowest energy is:

$$E_0 = \hbar\omega,$$

with degeneracy 1, corresponding to:

$$\phi_{0,0}(x,y) = \phi_0(x)\phi_0(y).$$

The next energy level is:

$$E_1 = 2\hbar\omega,$$

with degeneracy 2, corresponding to:

$$\phi_{1,0}(x,y) = \phi_1(x)\phi_0(y),$$
$$\phi_{0,1}(x,y) = \phi_0(x)\phi_1(y).$$

2.2 (4)

Since the two-dimensional harmonic oscillator is rotationally symmetric, its Hamiltonian is invariant under rotations around the z-axis.

Therefore:

$$\boxed{[L_z, H] = 0}.$$

2.3 (5)

First, we rewrite the angular momentum operator $L_z = xp_y - yp_x$ in terms of the creation and annihilation operators.

Recall that:

$$x_i = \sqrt{\frac{\hbar}{2m\omega}}(a_i^\dagger + a_i),$$
$$p_i = i\sqrt{\frac{m\hbar\omega}{2}}(a_i^\dagger - a_i).$$

Therefore:

$$\begin{aligned} L_z &= xp_y - yp_x \\ &= \frac{i\hbar}{2}\big[(a_x^\dagger + a_x)(a_y^\dagger - a_y) \\ &\qquad - (a_y^\dagger + a_y)(a_x^\dagger - a_x)\big]. \end{aligned}$$

Expanding the brackets:

$$L_z = \frac{i\hbar}{2}\big[a_x^\dagger a_y^\dagger - a_x^\dagger a_y + a_x a_y^\dagger - a_x a_y \\ - a_y^\dagger a_x^\dagger + a_y^\dagger a_x - a_y a_x^\dagger + a_y a_x\big].$$

Since the operators acting on different directions commute, we have:

$$[a_x, a_y] = [a_x, a_y^\dagger] = 0.$$

Thus, the expression can be simplified to:

$$\boxed{L_z = i\hbar(a_x a_y^\dagger - a_x^\dagger a_y)}.$$

The ground state of the two-dimensional oscillator is:

$$|0\rangle = |0\rangle_x |0\rangle_y.$$

Applying L_z to this state:

$$\begin{aligned} L_z|0\rangle_x|0\rangle_y &= i\hbar(a_x a_y^\dagger - a_x^\dagger a_y)|0\rangle_x|0\rangle_y \\ &= i\hbar\big(a_x|0\rangle_x a_y^\dagger|0\rangle_y - a_x^\dagger|0\rangle_x a_y|0\rangle_y\big) \\ &= 0. \end{aligned}$$

Therefore, the ground state is an eigenstate of L_z, with eigenvalue:

$$\boxed{L_z = 0}.$$

2.4 (6)

The first excited energy level is two-fold degenerate, with the two eigenstates:

$$|1\rangle_x|0\rangle_y, \qquad |0\rangle_x|1\rangle_y.$$

Using the expression of L_z obtained in (5), we first calculate:

$$\begin{aligned} L_z|0\rangle_x|1\rangle_y &= i\hbar(a_x a_y^\dagger - a_x^\dagger a_y)|0\rangle_x|1\rangle_y \\ &= i\hbar\big(a_x|0\rangle_x a_y^\dagger|1\rangle_y - a_x^\dagger|0\rangle_x a_y|1\rangle_y\big) \\ &= -i\hbar|1\rangle_x|0\rangle_y. \end{aligned}$$

Similarly:

$$\begin{aligned} L_z|1\rangle_x|0\rangle_y &= i\hbar(a_x a_y^\dagger - a_x^\dagger a_y)|1\rangle_x|0\rangle_y \\ &= i\hbar\big(a_x|1\rangle_x a_y^\dagger|0\rangle_y - a_x^\dagger|1\rangle_x a_y|0\rangle_y\big) \\ &= i\hbar|0\rangle_x|1\rangle_y. \end{aligned}$$

Now, consider a linear combination of these two states:

$$|\psi\rangle = a|1\rangle_x|0\rangle_y + b|0\rangle_x|1\rangle_y.$$

Applying L_z, we obtain:

$$L_z|\psi\rangle = i\hbar\big[a|0\rangle_x|1\rangle_y - b|1\rangle_x|0\rangle_y\big].$$

For this state to be an eigenstate of L_z, it must satisfy:

$$L_z|\psi\rangle = \ell|\psi\rangle.$$

Comparing the coefficients of the two independent states, we obtain:

$$\begin{cases} -i\hbar b = \ell a, \\ i\hbar a = \ell b. \end{cases}$$

Eliminating ℓ:

$$-\frac{b}{a} = \frac{a}{b},$$

which gives:

$$a^2 + b^2 = 0.$$

Therefore:

$$b = \pm ia.$$

Using the normalization condition:

$$|a|^2 + |b|^2 = 1,$$

we can choose:

$$a = \frac{1}{\sqrt{2}}, \qquad b = \pm\frac{i}{\sqrt{2}}.$$

Thus, the two eigenstates are:

$$\boxed{|\psi_\pm\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle_x|0\rangle_y \pm i|0\rangle_x|1\rangle_y\right)}.$$

Substituting these states into the eigenvalue equation, we obtain:

$$\boxed{L_z|\psi_\pm\rangle = \pm\hbar|\psi_\pm\rangle}.$$

3 III

3.1 (7)

Expanding the Hamiltonian:

$$H(B) = \frac{1}{2m}\left[\left(p_x - \frac{eB}{2}y\right)^2 + \left(p_y + \frac{eB}{2}x\right)^2\right] + \frac{1}{2}m\omega^2(x^2 + y^2).$$

Since p_x commutes with y, and p_y commutes with x, we can rearrange the cross terms:

$$H(B) = H(0) + \frac{eB}{2m}(xp_y - yp_x) + \frac{e^2B^2}{8m}(x^2 + y^2).$$

Comparing this with

$$H(B) = H(0) + W_1B + W_2B^2,$$

we obtain:

$$\boxed{W_1 = \frac{e}{2m}L_z}.$$

3.2 (8)

From the previous results:

$$W_1 = \frac{e}{2m} L_z.$$

Therefore:

$$W_1 |0\rangle_x |0\rangle_y = 0,$$

and

$$W_1 |\psi_\pm\rangle = \pm \frac{e\hbar}{2m} |\psi_\pm\rangle.$$

Using the definition of the magnetic moment:

$$\mu = - \left. \frac{\partial E(B)}{\partial B} \right|_{B=0},$$

we obtain:

$$\boxed{\mu_0 = 0, \qquad \mu_\pm = \mp \frac{e\hbar}{2m}}.$$

3.3 (9)

Combining the B^2 term with the harmonic oscillator potential, we can rewrite the Hamiltonian as:

$$H(B) = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2} m \Omega^2 (x^2 + y^2) + \frac{eB}{2m} L_z,$$

where:

$$\Omega = \sqrt{\omega^2 + \frac{e^2 B^2}{4m^2}}.$$

Thus, the exact energy eigenvalues are:

$$E(B) = \hbar\Omega(n_x + n_y + 1) + \frac{eB}{2m} l_z.$$

For the ground state:

$$\boxed{E_0(B) = \hbar\Omega}.$$

For the first excited states:

$$\boxed{E_{1,\pm}(B) = 2\hbar\Omega \pm \frac{e\hbar B}{2m}}.$$

Taking the limit $\omega \to 0$:

$$\boxed{E_0(B) \to \frac{e\hbar|B|}{2m}}$$

and

$$\boxed{E_{1,\pm}(B) \to \frac{e\hbar|B|}{m} \pm \frac{e\hbar B}{2m}}.$$